AWS CERTIFIED SOLUTIONS ARCHITECT · PROFESSIONAL

SAP-C02 Field Guide

Every in-scope service in one line, the look-alikes side by side, scenario keywords, and the numbers and lists worth memorizing.

01	**In-Scope Services**	One line per service, grouped by category
02	**Look-Alike Comparisons**	The distractor pairs, side by side
03	**Keyword Cheatsheet**	Scenario phrase to likely answer
04	**Numbers to Memorize**	Limits, quotas, and timings
05	**Lists to Memorize**	Engines, types, and options to know in full

How to read this guide

FOCUS — Heavily tested: themes the exam guide repeats across domains (security and IAM, networking, governance, cost, reliability and DR, databases, migration). Study these more deeply.

LEGACY — Still on the exam list, but closed to new customers, scheduled for retirement, or shut down. Know what it does, and expect newer questions to favour its replacement.

Based on the official SAP-C02 exam guide and AWS documentation · Current as of October 2026

PART 1

In-Scope Services

ANALYTICS

Service	One-liner
Amazon Athena	Serverless SQL directly on S3 data; pay per data scanned, so partition and use Parquet.
AWS Data Exchange	Find, subscribe to, and consume third-party datasets straight into S3, Redshift, or via API.
Amazon Data Firehose	Zero-admin delivery of streaming data to S3, Redshift, OpenSearch, Splunk, with buffering and transforms; near real time, not replayable.
Amazon EMR	Managed Hadoop/Spark/Hive/Presto clusters (EC2, EKS, or Serverless) for big data processing when you need framework control.
AWS Glue	Serverless ETL (Spark) plus the Data Catalog that Athena, EMR, and Redshift Spectrum share as a metastore.
Amazon Kinesis Data Streams	Real-time, ordered (per shard), replayable stream with multiple consumers; think "SQS you can rewind and fan out."
AWS Lake Formation	Governance layer over the Glue Catalog for fine-grained (table, column, row, cell) data lake permissions and cross-account sharing.
Amazon Managed Service for Apache Flink	Stateful real-time stream processing (windows, joins, aggregations) over Kinesis or MSK.
Amazon MSK	Managed Apache Kafka; pick it when the scenario already uses Kafka or needs Kafka APIs.
Amazon OpenSearch Service	Managed search and log analytics (full-text search, dashboards); common Firehose destination.
Amazon QuickSight	Serverless BI dashboards with row-level security; now branded Amazon Quick Sight inside Amazon Quick (formerly Quick Suite).

APPLICATION INTEGRATION

Service	One-liner
Amazon AppFlow	No-code data transfer between SaaS apps (Salesforce, SAP, Zendesk) and AWS services like S3 or Redshift.
AWS AppSync	Managed GraphQL (and real-time pub/sub) API layer over DynamoDB, Lambda, and other sources, with offline sync.
Amazon EventBridge	Serverless event bus with content-based routing, cross-account buses, SaaS sources, Pipes, and Scheduler.
Amazon MQ	Managed ActiveMQ/RabbitMQ; the lift-and-shift answer when apps use JMS, AMQP, MQTT, or STOMP.
Amazon SNS	Pub/sub fan-out to SQS, Lambda, HTTP, email, and SMS; pair with SQS for durable fan-out.
Amazon SQS	Fully managed queue for decoupling; Standard for throughput, FIFO for ordering and exactly-once processing.
AWS Step Functions	Serverless state machine orchestration with retries, branching, and human approval; Standard vs Express workflows.

BLOCKCHAIN

Service	One-liner
Amazon Managed Blockchain LEGACY	Managed Hyperledger Fabric and Ethereum networks for multi-party ledgers; in Sunset, end of support 29 Sep 2027.

BUSINESS APPLICATIONS

Service	One-liner
Amazon SES	Scalable transactional and marketing email sending and receiving.

CLOUD FINANCIAL MANAGEMENT FOCUS

Service	One-liner
AWS Budgets	Cost, usage, and RI/Savings Plans budgets with alerts and automated actions (for example, apply an SCP).
AWS Cost and Usage Report	Most granular billing data delivered to S3, queried with Athena or QuickSight for chargeback.
AWS Cost Explorer	Visualize and forecast spend, and get RI and Savings Plans purchase recommendations.
Savings Plans	1 or 3 year $/hour commitment; Compute SP flexes across EC2, Fargate, and Lambda, EC2 Instance SP is cheaper but narrower.

COMPUTE

Service	One-liner
AWS App Runner LEGACY	Source code or container to a running HTTPS web service with no infrastructure to manage; closed to new customers (Maintenance, Mar 2026).
AWS Auto Scaling	Unified scaling plans across EC2, ECS, DynamoDB, and Aurora resources.
AWS Batch	Managed batch job queues and scheduling on EC2, Spot, or Fargate.
AWS Elastic Beanstalk	PaaS that deploys web apps onto EC2, ELB, and Auto Scaling for you; supports blue/green via URL swap.
Amazon EC2	Virtual servers; know purchase options (On-Demand, Spot, RI, SP, Dedicated Hosts) and placement groups.
Amazon EC2 Auto Scaling	Scale EC2 fleets with target tracking, step, scheduled, or predictive policies; warm pools and lifecycle hooks.
AWS Fargate	Serverless compute engine for ECS and EKS tasks; no instances to patch.
AWS Lambda	Event-driven functions up to 15 minutes; know concurrency, provisioned concurrency, and VPC access.
Amazon Lightsail	Simple fixed-price VPS bundles for small apps; rarely the answer at enterprise scale.
AWS Outposts	AWS hardware in your data center for low latency or data residency with AWS APIs.
AWS Wavelength	Compute at 5G carrier edge for single-digit millisecond mobile latency.

CONTAINERS

Service	One-liner
Amazon ECR	Private container registry with image scanning and cross-Region/cross-account replication.
Amazon ECS	AWS-native container orchestration on EC2 or Fargate.

Service	One-liner
Amazon ECS Anywhere	Run ECS tasks on your on-premises servers, managed from the ECS control plane.
Amazon EKS	Managed Kubernetes control plane; pick when the scenario says Kubernetes or portability.
Amazon EKS Anywhere	Run and manage Kubernetes clusters on your own infrastructure using EKS tooling.
Amazon EKS Distro	The open-source Kubernetes distribution EKS uses, self-managed anywhere.

DATABASE FOCUS

Service	One-liner
Amazon Aurora	MySQL/PostgreSQL compatible, 6 copies across 3 AZs, up to 15 replicas, Global Database for cross-Region DR.
Amazon Aurora Serverless	Aurora that scales capacity automatically; fits spiky or unpredictable workloads.
Amazon DocumentDB	Managed MongoDB-compatible document database.
Amazon DynamoDB	Serverless key-value NoSQL at any scale; Global Tables, DAX, Streams, PITR, on-demand vs provisioned.
Amazon ElastiCache	Managed Redis/Valkey or Memcached for caching, sessions, and leaderboards.
Amazon Keyspaces	Serverless Apache Cassandra-compatible database.
Amazon Neptune	Managed graph database for relationships (social, fraud, knowledge graphs).
Amazon RDS	Managed relational engines (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2); Multi-AZ for HA, read replicas for scale.
Amazon Redshift	Columnar data warehouse; Spectrum queries S3, Serverless option, data sharing across accounts.
Amazon Timestream	Time-series database for IoT and operational metrics; Timestream for LiveAnalytics is closed to new customers, Timestream for InfluxDB is the active option.

DEVELOPER TOOLS

Service	One-liner
AWS CodeArtifact	Managed package repository (npm, Maven, PyPI) with upstream proxying.
AWS CodeBuild	Serverless build and test runner.
AWS CodeDeploy	Automated deployments to EC2, on-premises, Lambda, and ECS with in-place, blue/green, or canary.
Amazon CodeGuru LEGACY	ML-powered code review (Reviewer) and runtime profiling (Profiler); CodeGuru Reviewer is closed to new customers.
AWS CodePipeline	CI/CD orchestration chaining source, build, test, approval, and deploy stages.
AWS X-Ray	Distributed tracing to find latency and errors across microservices.

END USER COMPUTING

Service	One-liner
Amazon AppStream 2.0	Stream individual desktop applications to a browser; renamed Amazon WorkSpaces Applications in Nov 2025.
Amazon WorkSpaces	Managed persistent virtual desktops (DaaS).

FRONTEND WEB AND MOBILE

Service	One-liner
AWS Amplify	Build and host full-stack web and mobile apps with managed backends and CI/CD hosting.
Amazon API Gateway	Managed REST, HTTP, and WebSocket APIs with throttling, auth, caching, and private endpoints.
AWS Device Farm	Test apps on real mobile devices and browsers.
Amazon Pinpoint LEGACY	Multichannel customer engagement (campaigns, SMS, push); in Sunset, end of support 30 Oct 2026, features moving to AWS End User Messaging and Amazon Connect.

INTERNET OF THINGS (IOT)

Service	One-liner
AWS IoT Core	Managed MQTT broker and rules engine connecting devices to AWS services.
AWS IoT Device Defender	Audit IoT fleets for security misconfigurations; the Detect (anomaly monitoring) feature is closed to new customers.
AWS IoT Device Management	Onboard, organize, monitor, and remotely update devices at scale (OTA jobs).
AWS IoT Events LEGACY	Detected state changes in sensor data; fully shut down 20 May 2026.
AWS IoT Greengrass	Run Lambda, containers, and ML inference on edge devices, even offline; use V2 (V1 reached end of support 7 Oct 2026).
AWS IoT SiteWise	Collect and model industrial equipment data at scale.
AWS IoT Things Graph LEGACY	Visual tool to connect devices into workflows; fully shut down 9 Nov 2022.
AWS IoT 1-Click LEGACY	Simple devices that triggered Lambda with one click; fully shut down 29 Jan 2025.

MACHINE LEARNING

Service	One-liner
Amazon Comprehend	NLP for sentiment, entities, key phrases, language, and PII detection in text.
Amazon Fraud Detector LEGACY	Managed fraud detection models for payments and fake accounts; closed to new customers since Nov 2025.
Amazon Kendra LEGACY	ML-powered enterprise search across documents; closed to new customers (Maintenance, Jun 2026).
Amazon Lex	Conversational chatbots with speech and text (the engine behind Alexa).
Amazon Personalize	Real-time personalized recommendations without ML expertise.
Amazon Polly	Text to lifelike speech.
Amazon Rekognition	Image and video analysis: objects, faces, moderation, text.
Amazon SageMaker AI	Build, train, and deploy custom ML models end to end.
Amazon Textract	Extract text, forms, and tables from scanned documents (beyond plain OCR).
Amazon Transcribe	Speech to text, including call analytics and PII redaction.
Amazon Translate	Neural machine translation between languages.

MEDIA SERVICES

Service	One-liner
Amazon Elastic Transcoder LEGACY	Legacy media transcoding; fully shut down 13 Nov 2025, replaced by AWS Elemental MediaConvert.
Amazon Kinesis Video Streams	Ingest, store, and process video streams from cameras for playback and analytics.

MANAGEMENT AND GOVERNANCE FOCUS

Service	One-liner
AWS CloudFormation	Infrastructure as code; StackSets deploy across accounts and Regions in an Organization.
AWS CloudTrail	Records API calls for audit; use an organization trail to a central, locked-down log archive bucket.
Amazon CloudWatch	Metrics, alarms, dashboards, and cross-account observability.
Amazon CloudWatch Logs	Centralized log storage, Logs Insights queries, subscription filters to Kinesis or Lambda.
AWS CLI	Command line control of AWS services and scripting.
AWS Compute Optimizer	ML rightsizing recommendations for EC2, ASG, EBS, Lambda, and ECS on Fargate.
AWS Config	Records resource configuration history and evaluates compliance rules with auto-remediation; aggregators for multi-account.
AWS Control Tower	Automated multi-account landing zone with guardrails (SCPs and Config rules) and Account Factory.
AWS Health Dashboard	Personalized view of AWS events affecting your resources; organizational view across accounts.
AWS License Manager	Track and enforce software license usage (BYOL) across accounts.
Amazon Managed Grafana	Managed Grafana dashboards over multiple data sources.
Amazon Managed Service for Prometheus	Managed Prometheus-compatible metrics monitoring for containers.
AWS Management Console	Web UI for AWS.
AWS Organizations	Multi-account management: OUs, consolidated billing, SCPs, RCPs, and delegated administrators.
AWS Proton LEGACY	Platform teams published standard templates for self-service deployment; reached end of support 7 Oct 2026.
AWS Service Catalog	Curated, approved CloudFormation products end users can launch without broad IAM permissions.
Service Quotas	View and request increases to service limits centrally, including quota templates for new accounts.
AWS Systems Manager	Ops hub for EC2 and on-prem: Session Manager, Patch Manager, Run Command, Parameter Store, Automation.
AWS Trusted Advisor	Best-practice checks across cost, security, fault tolerance, performance, and quotas.
AWS Well-Architected Tool	Review workloads against the six Well-Architected pillars and track improvement plans.

MIGRATION AND TRANSFER FOCUS

Service	One-liner
AWS Application Discovery Service LEGACY	Discover on-prem servers, utilization, and dependencies to plan migrations; closed to new customers (Maintenance, Oct 2025), and its Discovery Connector is shut down.

Service	One-liner
AWS Application Migration Service (MGN)	Block-level replication for lift-and-shift rehosting of servers to EC2 with minimal downtime.
AWS DMS	Migrate and continuously replicate (CDC) databases with minimal downtime, homogeneous or heterogeneous.
AWS DataSync	Online, accelerated file and object transfer between on-prem (NFS, SMB, HDFS) and S3, EFS, or FSx.
AWS Migration Hub LEGACY	Single place to track migration progress across tools; closed to new customers (Maintenance, Nov 2025).
AWS SCT	Converts database schema and code between engines (for example, Oracle to Aurora PostgreSQL); pair with DMS.
AWS Snow Family LEGACY	Physical devices for offline bulk transfer and edge compute; Snowball Edge is existing customers only, Snowcone and Snowmobile are gone (see Part 4).
AWS Transfer Family	Managed SFTP, FTPS, FTP, and AS2 endpoints backed by S3 or EFS.

NETWORKING AND CONTENT DELIVERY FOCUS

Service	One-liner
Amazon CloudFront	Global CDN with edge caching, OAC for private S3, Lambda@Edge and CloudFront Functions.
AWS Direct Connect	Dedicated private link from on-prem to AWS; consistent latency, use VPN backup or dual DX for resiliency.
Elastic Load Balancing	ALB (L7, path/host routing), NLB (L4, static IP, extreme performance), GWLB (inline appliances).
AWS Global Accelerator	Anycast static IPs routing over the AWS backbone for fast, non-HTTP-friendly multi-Region failover.
AWS PrivateLink	Expose a service privately to other VPCs or accounts via interface endpoints, without peering or overlapping CIDR issues.
Amazon Route 53	DNS with routing policies (latency, failover, weighted, geolocation), health checks, and Resolver for hybrid DNS.
AWS Transit Gateway	Hub-and-spoke router for thousands of VPCs and on-prem networks; share across accounts with RAM.
Amazon VPC	Isolated network: subnets, route tables, SGs (stateful), NACLs (stateless), endpoints, peering (non-transitive).
AWS VPN	Encrypted IPsec Site-to-Site VPN over the internet, plus Client VPN for remote users.

SECURITY, IDENTITY, AND COMPLIANCE FOCUS

Service	One-liner
AWS Artifact	Download AWS compliance reports (SOC, PCI, ISO) and accept agreements like the BAA.
AWS Audit Manager LEGACY	Continuously collect evidence and map it to compliance frameworks; closed to new customers (Maintenance, Mar 2026).
AWS Certificate Manager (ACM)	Free public TLS certificates with auto-renewal for ELB, CloudFront, and API Gateway; private CA option.
AWS CloudHSM	Single-tenant, FIPS 140 validated HSMs where you exclusively control the keys.
Amazon Cognito	User pools (sign-up and sign-in for app users) and identity pools (temporary AWS credentials).
Amazon Detective	Investigate root cause of security findings using graph analysis of logs.

Service	One-liner
AWS Directory Service	Managed Microsoft AD or AD Connector (proxy to on-prem AD); Simple AD is closed to new customers.
AWS Firewall Manager	Centrally deploy WAF, Shield Advanced, SG, and Network Firewall policies across an Organization.
Amazon GuardDuty	Managed threat detection from CloudTrail, VPC Flow Logs, DNS logs, and more; delegated admin for the org.
AWS IAM Identity Center	Central workforce SSO and permission sets across all Organization accounts, federated to your IdP.
AWS IAM	Users, roles, and policies; know evaluation logic: explicit deny beats allow, plus SCPs, boundaries, and session policies.
Amazon Inspector	Automated vulnerability scanning for EC2, ECR images, and Lambda.
AWS KMS	Managed encryption keys; key policies, grants, multi-Region keys, and cross-account key sharing.
Amazon Macie	ML discovery of sensitive data (PII) in S3.
AWS Network Firewall	Managed stateful network firewall and IPS for VPC traffic inspection, often in a centralized inspection VPC.
AWS RAM	Share resources (TGW, subnets, Resolver rules) across accounts or the Organization.
AWS Secrets Manager	Store and automatically rotate secrets like DB credentials; choose over Parameter Store when rotation is needed.
AWS Security Hub	Aggregate and prioritize findings plus run security standard checks across accounts and Regions.
AWS STS	Issues temporary credentials via AssumeRole, the backbone of cross-account access and federation.
AWS Shield	DDoS protection; Standard is free, Advanced adds L7 response team support and cost protection.
AWS WAF	Layer 7 filtering (SQLi, XSS, rate limits, geo) on CloudFront, ALB, API Gateway, and AppSync.

STORAGE

Service	One-liner
AWS Backup	Centralized, policy-based backups across services and accounts, with Vault Lock and cross-Region copy.
Amazon EBS	Block storage for EC2 in one AZ; gp3, io2, snapshots, and Fast Snapshot Restore.
AWS Elastic Disaster Recovery	Continuous block replication for low RPO/RTO DR of servers into AWS.
Amazon EFS	Elastic, multi-AZ NFS file system for Linux, shared across many instances.
Amazon FSx	Managed Windows File Server, Lustre (HPC), NetApp ONTAP, and OpenZFS file systems.
Amazon S3	Durable object storage; storage classes, lifecycle, replication (CRR/SRR), Object Lock, access points.
Amazon S3 Glacier	Low-cost archive storage classes (Instant, Flexible, Deep Archive); the old vault-based Glacier API is closed to new customers, so use the S3 storage classes.
AWS Storage Gateway	Hybrid access to cloud storage: S3 File, FSx File, Volume, and Tape Gateway.

PART 2

Look-Alike Comparisons

These are the pairs and triples the exam deliberately puts side by side as distractors.

Cognito User Pool vs Identity Pool

	User Pool	Identity Pool
Answers	"Who are you?" (authentication)	"What AWS resources can you touch?" (authorization)
Output	JWT tokens (ID, access, refresh)	Temporary AWS credentials via STS
Use for	Sign-up/sign-in, social and SAML login for app users	Letting app users call S3 or DynamoDB directly
Together	User Pool logs the user in, Identity Pool exchanges that token for an IAM role	

SCP vs IAM Policy vs Permission Boundary vs Resource Policy FOCUS

	Grants access?	Scope	Exam cue
SCP	No, only sets the maximum	Every principal in an account or OU (not the management account)	"Prevent any account from leaving a Region / disabling CloudTrail"
RCP	No, only sets the maximum	Resources in member accounts	"Data perimeter: no external principal can access our S3 or KMS"
Permission boundary	No, only sets the maximum	One IAM user or role	"Let developers create roles without escalating privilege"
Identity policy	Yes	Attached to user, group, or role	Normal permissions
Resource policy	Yes	Attached to resource (S3, KMS, SQS)	Cross-account access without assuming a role

Evaluation order: explicit Deny anywhere wins, then every applicable limiter (SCP, RCP, boundary, session policy) must allow, then an identity or resource policy must allow.

Secrets Manager vs Systems Manager Parameter Store FOCUS

	Secrets Manager	Parameter Store
Built-in rotation	Yes (native for RDS, Redshift, DocumentDB)	No (DIY with Lambda)
Cost	Per secret per month	Standard tier free
Cross-Region replication	Yes	No
Pick when	"Rotate DB credentials automatically"	"Store config values cheaply"

Kinesis Data Streams vs Data Firehose vs SQS vs MSK

	Kinesis Data Streams	Data Firehose	SQS	MSK
Latency	Real time (~200 ms)	Near real time (buffered)	Real time	Real time
Replay	Yes (retention)	No	No (deleted on consume)	Yes
Consumers	Many, each reads all	Delivers to a destination	Competing consumers, one gets each message	Many

	Kinesis Data Streams	Data Firehose	SQS	MSK
Ops effort	Shards or on-demand	Zero	Zero	Brokers to size
Exam cue	"Multiple apps process the same stream"	"Load streaming data into S3/Redshift with no code"	"Decouple, buffer work"	"Existing Kafka"

SNS vs SQS vs EventBridge

	SNS	SQS	EventBridge
Model	Push pub/sub	Pull queue	Event bus with rules
Filtering	Message attribute filter	None	Rich content-based patterns
Pick when	High-throughput fan-out	Buffering and retries	SaaS or AWS service events, cross-account routing, schedules

VPC Peering vs Transit Gateway vs PrivateLink vs Cloud WAN FOCUS

	VPC Peering	Transit Gateway	PrivateLink	Cloud WAN
Topology	1:1, non-transitive	Hub and spoke, transitive	Service consumer to provider	Global managed WAN
Overlapping CIDRs	Not allowed	Not allowed	Allowed	Not allowed
Cost	No hourly fee, cheapest data path	Hourly per attachment plus per GB	Per endpoint hour plus per GB	Highest
Exam cue	"Few VPCs, lowest cost"	"Hundreds of VPCs plus on-prem"	"Expose one service to many accounts / overlapping IPs"	"Multi-Region global network with central policy"

Gateway Endpoint vs Interface Endpoint FOCUS

	Gateway Endpoint	Interface Endpoint
Services	S3 and DynamoDB only	Most AWS services and PrivateLink services
Mechanism	Route table entry	ENI with private IP in your subnet
Cost	Free	Hourly plus per GB
Reachable from on-prem	No	Yes (via DX or VPN)

Site-to-Site VPN vs Direct Connect FOCUS

	Site-to-Site VPN	Direct Connect
Path	Encrypted IPsec over internet	Private dedicated fibre
Setup time	Minutes	Weeks to months
Bandwidth	Up to 1.25 Gbps per tunnel (ECMP to aggregate on TGW)	1, 10, 100, 400 Gbps dedicated; hosted from 50 Mbps
Encryption	Built in	Not by default, add MACsec (10/100/400 Gbps, select locations) or VPN over DX
Exam cue	"Quick, cheap, temporary or backup"	"Consistent latency, high throughput, large daily data"

ALB vs NLB vs GWLB

	ALB	NLB	GWLB
Layer	7 (HTTP/HTTPS/gRPC)	4 (TCP/UDP/TLS)	3 (GENEVE)
Key feature	Path, host, header routing, OIDC auth, WAF	Static or Elastic IP, millions of rps, ultra-low latency, PrivateLink provider	Inserts third-party firewalls and IDS transparently

CloudFront vs Global Accelerator FOCUS

	CloudFront	Global Accelerator
Caches content	Yes	No
Protocols	HTTP/HTTPS	TCP/UDP (any)
Entry point	DNS name	2 static anycast IPs
Exam cue	"Static and dynamic web content, edge caching"	"Gaming, VoIP, IoT; static IP; fast Regional failover"

DataSync vs Storage Gateway vs Transfer Family vs Snowball FOCUS

	DataSync	Storage Gateway	Transfer Family	Snowball Edge
Purpose	One-time or scheduled bulk copy	Ongoing hybrid access with local cache	SFTP/FTPS/AS2 endpoint for partners	Offline transfer when network is too slow
Exam cue	"Migrate NFS/SMB/HDFS to S3/EFS/FSx"	"On-prem apps keep using file, volume, or tape interface"	"Partners upload via SFTP, no server changes for them"	"Limited bandwidth, hundreds of TB" (existing customers only, see numbers)

MGN vs DMS vs SCT vs Elastic Disaster Recovery FOCUS

	MGN	DMS	SCT	DRS
Moves	Whole servers (rehost)	Database data, with CDC	Schema and stored procedures	Servers for DR, not migration
Exam cue	"Lift and shift 500 VMs, minimal downtime"	"Migrate DB with minimal downtime"	"Oracle to Aurora PostgreSQL (heterogeneous)"	"Low RPO/RTO DR for on-prem servers"

DR Strategies FOCUS

Strategy	RPO/RTO	Cost	What runs in DR Region
Backup and restore	Hours	$	Only backups
Pilot light	Tens of minutes	$$	Data replicated, core infra off
Warm standby	Minutes	$$$	Scaled-down full stack running
Multi-site active/active	Near zero	$$$$	Full production in both

Aurora Global Database vs RDS Cross-Region Read Replica vs DynamoDB Global Tables FOCUS

	Aurora Global DB	RDS Cross-Region Replica	DynamoDB Global Tables
Replication	Storage-level, typically under 1 s lag, up to 10 secondary Regions	Async engine-level	Multi-active, all Regions writable

	Aurora Global DB	RDS Cross-Region Replica	DynamoDB Global Tables
Failover	Managed, about a minute	Manual promotion	No failover needed
Exam cue	"Relational, lowest RPO across Regions"	"Cheaper, tolerates more lag"	"Multi-Region writes, NoSQL"

EFS vs FSx Family vs EBS

	EBS	EFS	FSx Windows	FSx Lustre	FSx ONTAP
Protocol	Block	NFS	SMB, AD integrated	Lustre (HPC)	NFS, SMB, iSCSI
Shared	Single instance (Multi-Attach io1/io2 only)	Thousands of Linux clients	Windows clients	HPC clusters, S3-linked	Multi-protocol, NetApp features

C vs M vs R vs T Instances

	C (compute optimized)	M (general purpose)	R (memory optimized)	T (burstable)
Memory per vCPU	2 GiB (c7i.xlarge: 4 vCPU, 8 GiB)	4 GiB	8 GiB (r7i.xlarge: 4 vCPU, 32 GiB)	Small sizes from 0.5 GiB
CPU model	Sustained high CPU	Balanced, sustained	Sustained, memory-bound	Baseline plus burst on CPU credits
Use for	Batch, media transcoding, game servers, HPC, CPU-bound web tiers	Web and app servers, mixed workloads, small to medium DBs	In-memory DBs and caches, large relational DBs, real-time analytics	Dev/test, low-traffic web, microservices that idle most of the day
Watch out	Low memory per vCPU	None, default choice	Higher cost per vCPU	T3/T3a/T4g/T8i launch in unlimited mode by default: sustained CPU above baseline is charged as surplus credits, so use M or C instead
Exam cue	"CPU-bound, compute-intensive"	"Balanced, no clear bottleneck"	"Large dataset in memory, out of memory errors"	"Low average CPU with occasional spikes, lowest cost"

Need even more memory per vCPU (16 to 32 GiB, e.g. SAP HANA)? Use X or U (high memory). Name suffixes: g = Graviton, a = AMD, i = Intel, d = instance store, n = network optimized.

Redis OSS (or Valkey) vs Memcached FOCUS

	Redis OSS / Valkey	Memcached
Data types	Strings, hashes, lists, sets, sorted sets, bitmaps, HyperLogLog, geospatial	Simple strings and objects
Multithreaded	No	Yes, uses many cores per node
Replication and failover	Yes, replicas with Multi-AZ automatic failover	No, losing a node loses its data
Backup and restore	Yes (snapshots)	Serverless caches only, not node-based clusters
Pub/Sub, sorted sets, geospatial	Yes	No
Multi-Region	Global Datastore	No

	Redis OSS / Valkey	Memcached
Scaling	Cluster mode enabled for sharding and online resharding	Add or remove nodes, client-side partitioning
Pick when	Leaderboards (sorted sets), session store that must survive failure, pub/sub, HA cache	Simplest cache, large multithreaded nodes, scale out and in freely, loss of cached data is acceptable

Valkey is the Redis OSS-compatible engine getting new features: vector search from 8.2, full-text search and Multi-AZ durability from 9.0.

Security Detection Services FOCUS

Service	Looks at	One-word job
GuardDuty	CloudTrail, VPC Flow Logs, DNS logs, EKS, S3 events	Threats
Inspector	EC2, ECR images, Lambda code	Vulnerabilities (CVEs)
Macie	S3 objects	Sensitive data (PII)
Detective	Findings plus logs as a graph	Investigation
Security Hub	Findings from all of the above plus standards	Aggregation
Config	Resource configuration	Compliance
CloudTrail	API calls	Audit

Shield Standard vs Shield Advanced FOCUS

	Standard	Advanced
Cost	Free, automatic	Monthly subscription per org
Layers	3 and 4	3, 4, and 7 (with WAF)
Extras	None	Shield Response Team, DDoS cost protection, health-based detection

Organizations vs Control Tower FOCUS

Organizations is the engine (accounts, OUs, SCPs, consolidated billing). Control Tower is the opinionated landing zone built on top (Account Factory, log archive and audit accounts, preventive and detective controls). Exam cue: "set up a governed multi-account environment quickly with best practices" means Control Tower.

Savings Plans vs Reserved Instances vs Spot FOCUS

	Compute SP	EC2 Instance SP	Standard RI	Convertible RI	Spot
Flexibility	Any family, Region, OS, Fargate, Lambda	Any size and OS within one family and Region	Fixed attributes	Can exchange	Can be interrupted
Discount	Up to 66%	Up to 72%	Up to 72%	Lower than Standard (sources quote 54% to 66%)	Up to 90%
Exam cue	"Steady spend, changing tech"	"Committed to one family"	"Need capacity reservation (zonal)"	"Might change family"	"Fault-tolerant, stateless, batch"

PART 3

Keyword Cheatsheet

Hybrid connectivity FOCUS

If the scenario says	Think
Connect on-prem quickly, encrypted, low cost	Site-to-Site VPN
Consistent latency, dedicated bandwidth, large data transfers	Direct Connect
DX must be encrypted	MACsec (10/100/400 Gbps dedicated, select locations) or IPsec VPN over DX
DX highly resilient	Two DX connections at two different DX locations ("maximum resiliency": 4 connections, 2 locations)
Cost-effective DX backup	Site-to-Site VPN as backup to DX
On-prem to many VPCs in many Regions over DX	Direct Connect Gateway plus Transit Gateway (transit VIF)
VPN bandwidth above 1.25 Gbps	Multiple VPN tunnels with ECMP on Transit Gateway
On-prem DNS resolves AWS private names (or reverse)	Route 53 Resolver inbound / outbound endpoints and forwarding rules
Share Resolver rules across accounts	AWS RAM
Branch offices connect to each other via AWS	VPN CloudHub (VGW) or Transit Gateway

Multi-account and governance FOCUS

If the scenario says	Think
Prevent all accounts from doing X	SCP
No one outside our org can access our resources	Data perimeter: RCPs, `aws:PrincipalOrgID`, VPC endpoint policies
New accounts automatically get baseline guardrails	Control Tower Account Factory (or Account Factory for Terraform)
Single sign-on to all accounts with corporate IdP	IAM Identity Center with SAML/SCIM
Developers create roles but cannot escalate	Permission boundaries
Access another account's resources	Cross-account IAM role with `sts:AssumeRole` (plus External ID for third parties)
Deploy stack to every account in an OU	CloudFormation StackSets (service-managed)
Central security tooling	Delegated administrator for GuardDuty, Security Hub, Config in a security account
Share subnets, TGW, or License configs across accounts	AWS RAM
Chargeback per team	Cost allocation tags, CUR plus Athena, AWS Budgets
Users only launch approved products	Service Catalog

Data and analytics

If the scenario says	Think
Query data in S3 with SQL, no infrastructure	Athena (plus Glue Data Catalog)
Column or row level permissions on the data lake	Lake Formation
Ingest clickstream into S3 with no code	Data Firehose
Real-time analytics over a stream with windows	Managed Service for Apache Flink
Existing Hadoop/Spark jobs	EMR
Data warehouse, complex joins, BI	Redshift (Spectrum for S3 data)
Discover schema automatically	Glue crawler
Log search and dashboards	OpenSearch Service
BI dashboards for business users	QuickSight

Migration FOCUS

If the scenario says	Think
Discover servers and dependencies first	Application Discovery Service (classic exam answer; closed to new customers since Oct 2025, AWS now points to AWS Transform)
Rehost with minimal downtime	Application Migration Service (MGN)
Database with minimal downtime	DMS with CDC (plus SCT if engines differ)
Large file shares, network available	DataSync
Too much data for the network	Snowball Edge for existing customers; new customers use Data Transfer Terminal or partners
Keep on-prem apps, back with S3	Storage Gateway
Retire mainframe or legacy batch	Modernize to containers or serverless (Batch, Step Functions)
VMware workloads with no change	VMware Cloud on AWS, or MGN

Resilience and performance FOCUS

If the scenario says	Think
Static IPs for global users, fast failover	Global Accelerator
DNS failover between Regions	Route 53 failover routing plus health checks
Reduce DB read load	Read replicas or ElastiCache
Sudden traffic spikes on relational DB	Aurora Serverless v2, RDS Proxy for connection storms
Lambda exhausting DB connections	RDS Proxy
Decouple and absorb spikes	SQS between tiers
Sessions survive instance loss	ElastiCache or DynamoDB for session state
Spiky unknown traffic on DynamoDB	On-demand capacity mode

If the scenario says	Think
Microsecond DynamoDB reads	DAX

Security FOCUS

If the scenario says	Think
Customer must control and own HSM keys, FIPS 140 Level 3	CloudHSM (or KMS custom key store)
Encrypt at rest with audit of key use	KMS customer managed key plus CloudTrail
Share encrypted snapshot or AMI cross-account	Re-encrypt with a CMK and grant the other account in the key policy
Block SQLi, XSS, rate limit by IP	WAF
Centrally manage WAF rules across accounts	Firewall Manager
Inspect east-west and egress traffic	Network Firewall in a centralized inspection VPC
Find PII in S3	Macie
SSH/RDP without bastions or open ports	Systems Manager Session Manager
Rotate database passwords	Secrets Manager
Detect compromised instances	GuardDuty
Auto-remediate non-compliant resources	Config rule plus SSM Automation remediation
Immutable backups, ransomware protection	AWS Backup Vault Lock, S3 Object Lock (compliance mode)

Cost FOCUS

If the scenario says	Think
Rightsize EC2, Lambda, EBS	Compute Optimizer
Steady usage, flexible families	Compute Savings Plans
Interruptible batch	Spot (with capacity-optimized allocation)
Infrequently accessed S3 with unknown pattern	S3 Intelligent-Tiering
Reduce NAT Gateway cost to S3	S3 Gateway endpoint
Cross-AZ data transfer cost high	Keep chatty components in same AZ, or use VPC endpoints
Alert or block on overspend	AWS Budgets with Budget Actions

PART 4

Numbers to Memorize

Verified 13 Oct 2026 against AWS documentation. Rows marked ✓ were checked against AWS documentation or AWS announcements; the few unmarked values are stable, long-standing figures. Exam questions may lag behind recent limit increases, so know both old and new values where shown.

Snow Family (status changed: read this first) FOCUS ✓

Device	Status	Key specs
Snowcone (8 TB HDD / 14 TB SSD)	Discontinued Nov 2024, support ended Nov 2025	Older exam questions may still mention it
Snowmobile (100 PB)	Retired March 2024	Classic "exabyte-scale" answer, now gone
Snowball Edge Storage Optimized 210 TB	Existing customers only since 7 Nov 2025	210 TB NVMe, up to 1.5 GB/s transfer
Previous generation (80 TB, 52 vCPU, GPU)	Shut down 12 Nov 2025	Often quoted in older practice exams
Snowball Edge Compute Optimized	Existing customers only since 7 Nov 2025	104 vCPUs, 416 GB RAM, 28 TB NVMe
Replacements	New customers	DataSync (online), AWS Data Transfer Terminal (physical), Outposts (edge compute)

Rule of thumb for "network vs device": 1 Gbps moves roughly 10 TB per day at full utilization, so 100 TB takes about 10 days on a saturated 1 Gbps link.

Networking FOCUS ✓

Item	Number
Site-to-Site VPN per tunnel	Up to 1.25 Gbps, 2 tunnels per connection
Transit Gateway VPC attachment	Up to 100 Gbps per AZ (older material says 50 Gbps)
Transit Gateway attachments	5,000 per TGW
TGW Connect (GRE) peer	Up to 5 Gbps, 4 peers per attachment (20 Gbps)
Direct Connect dedicated ✓	1, 10, 100, 400 Gbps (400 Gbps at select locations)
Direct Connect hosted ✓	50 Mbps to 25 Gbps (one VIF per hosted connection)
Direct Connect LAG ✓	Up to 4 links below 100 Gbps, or 2 links at 100/400 Gbps
VPC and subnet CIDR size ✓	/16 to /28
Reserved IPs per subnet ✓	5 (first four plus the last)
Security group rules ✓	60 inbound plus 60 outbound by default (adjustable)

Compute and serverless ✓

Item	Number
Lambda timeout	900 s (15 min), cannot be raised
Lambda memory	128 MB to 10,240 MB
Lambda /tmp	512 MB to 10,240 MB

Item	Number
Lambda sync payload	6 MB request and response
Lambda async payload	1 MB (was 256 KB)
Lambda scaling	1,000 execution environments per 10 s per function
API Gateway integration timeout	29 s default; Regional and private REST APIs can request more (up to 5 min)
Step Functions Standard ✓	Up to 1 year duration, exactly-once
Step Functions Express ✓	Up to 5 minutes, at-least-once (async)

Messaging and streaming

Item	Number
SQS message size ✓	1 MiB (was 256 KiB until Aug 2025; Extended Client Library with S3 for larger)
SQS retention ✓	Up to 14 days (default 4 days)
SQS visibility timeout ✓	Up to 12 h (default 30 s)
SQS delay / long poll	Delay up to 15 min ✓; long poll up to 20 s
SQS FIFO ✓	300 TPS per API action, 3,000 msg/s with batching; high-throughput mode up to 70,000 TPS in the largest Regions
Kinesis shard ✓	1 MB/s or 1,000 records/s in, 2 MB/s out (enhanced fan-out 2 MB/s per consumer)
Kinesis retention ✓	24 h default, up to 365 days
Firehose buffer	Delivers on size or time interval, whichever first (near real time)

Storage

Item	Number
S3 object max ✓	50 TB since Dec 2025 (was 5 TB; older questions use 5 TB); single PUT 5 GB; use multipart above 100 MB
S3 request rate ✓	At least 3,500 PUT/COPY/POST/DELETE and 5,500 GET/HEAD per second per prefix
S3 durability	11 nines
Glacier Instant Retrieval ✓	Milliseconds
Glacier Flexible Retrieval ✓	Expedited 1 to 5 min, Standard 3 to 5 h, Bulk 5 to 12 h (free)
Glacier Deep Archive ✓	Standard within 12 h, Bulk within 48 h
Minimum storage duration ✓	Standard-IA / One Zone-IA 30 days, Glacier IR and Flexible 90 days, Deep Archive 180 days
EBS io2 Block Express ✓	Up to 256,000 IOPS, 4,000 MB/s, 64 TiB; Multi-Attach to up to 16 Nitro instances
EBS gp3 ✓	3,000 IOPS and 125 MiB/s baseline at any size; now up to 80,000 IOPS and 2,000 MB/s (older material says 16,000 and 1,000)

Database FOCUS

Item	Number
Aurora ✓	6 copies across 3 AZs, up to 15 replicas
Aurora Global Database ✓	Typically under 1 s lag, up to 10 secondary Regions (was 5 until May 2025); primary readers plus secondary clusters cannot exceed 15

Item	Number
RDS Multi-AZ	Synchronous standby, automatic failover typically 60 to 120 s (Multi-AZ DB cluster: typically under 35 s)
RDS read replicas ✓	Up to 15 per source (MySQL, MariaDB, PostgreSQL); 5 for Oracle and SQL Server; 3 for Db2
DynamoDB item size ✓	400 KB (attribute names count)
DynamoDB PITR ✓	1 to 35 days, configurable per table (default 35)
RDS / Aurora automated backups ✓	Up to 35 days retention
ElastiCache	Redis/Valkey for persistence and replication; Memcached for simple multithreaded cache

Identity and security FOCUS

Item	Number
IAM role session ✓	15 min to the role's maximum (1 h to 12 h); role chaining capped at 1 h
SCPs ✓	Up to 10 attached per root, OU, or account; 10,240 characters each (raised from 5 and 5,120 in May 2026, so older questions use the old values)
KMS Encrypt API ✓	Up to 4,096 bytes with a symmetric key; larger data uses envelope encryption with data keys
KMS key deletion ✓	7 to 30 day waiting period (default 30)
CloudTrail ✓	90 days of management events in free Event history; use a trail to S3 for longer retention
Config	Continuous recording; aggregator for multi-account, multi-Region view

PART 5

Lists to Memorize

Lists the exam expects you to know in full. Checked 14 Oct 2026 against AWS documentation.

Databases FOCUS

List	Members	Exam cue
RDS engines	Db2, MariaDB, MySQL, Oracle, PostgreSQL, SQL Server (plus Aurora)	Engine not listed (e.g. Sybase, Informix)? Then EC2 self-managed, or migrate with DMS/SCT
RDS Custom engines	Oracle and SQL Server only	"Need OS or DB admin access on a managed DB"
Aurora compatibility	MySQL and PostgreSQL only	"Oracle to Aurora" means heterogeneous: SCT plus DMS
Aurora variants	Provisioned, Serverless v2, Global Database, PostgreSQL Limitless (sharding), Aurora DSQL (serverless, distributed, PostgreSQL-compatible, active-active multi-Region)	"Multi-Region active-active relational with strong consistency" points to DSQL
ElastiCache engines	Valkey, Redis OSS, Memcached	Replication, persistence, pub/sub: Valkey/Redis OSS. Simple multithreaded cache: Memcached
Graph	Neptune	Social graphs, fraud rings, recommendations
Document	DocumentDB	MongoDB-compatible JSON documents
Wide-column	Keyspaces	Cassandra-compatible, serverless
Time series	Timestream for InfluxDB	Timestream for LiveAnalytics is closed to new customers (June 2025)
Durable in-memory primary DB	MemoryDB	Microsecond reads, Multi-AZ durable; not just a cache
Ledger	QLDB LEGACY	End of support 31 Jul 2025; AWS points to Aurora PostgreSQL

Storage

List	Members	Exam cue
S3 storage classes	Standard, Express One Zone, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant Retrieval, Glacier Flexible Retrieval, Glacier Deep Archive, Outposts	Express One Zone: single-digit ms, one AZ, directory buckets
EBS volume types	gp3, gp2 (SSD); io2 Block Express, io1 (provisioned IOPS SSD); st1, sc1 (HDD); Magnetic (previous generation)	st1/sc1 cannot be boot volumes; Multi-Attach is io1/io2 only
FSx types	Windows File Server, Lustre, NetApp ONTAP, OpenZFS	SMB plus AD: Windows. HPC plus S3: Lustre. Multi-protocol: ONTAP. Linux NFS migration: OpenZFS
Storage Gateway types	S3 File, FSx File LEGACY, Volume (cached or stored), Tape	FSx File Gateway is closed to new customers. Cached: primary data in S3. Stored: primary data on-prem

Networking FOCUS

List	Members	Exam cue
Gateway endpoints	S3 and DynamoDB only	Everything else needs an interface endpoint
Route 53 routing policies	Simple, Weighted, Latency, Failover, Geolocation, Geoproximity, Multivalue answer, IP-based	Geolocation: by user location (compliance). Geoproximity: by resource location with bias. IP-based: by client CIDR
Direct Connect VIFs	Private (one VPC via VGW), Public (AWS public endpoints), Transit (TGW via DX Gateway)	"Many VPCs over one DX" means transit VIF
Load balancers	ALB, NLB, GWLB; Classic is previous generation	Classic in an answer is usually the wrong choice

Compute and deployment

List	Members	Exam cue
EC2 placement groups	Cluster, Partition (up to 7 per AZ), Spread (up to 7 instances per AZ), Precision time (new)	Cluster: low latency HPC. Partition: HDFS, Cassandra, Kafka. Spread: few critical instances
CodeDeploy on EC2	In-place or blue/green; AllAtOnce, HalfAtATime, OneAtATime (default)	Blue/green on EC2 replaces instances behind a load balancer
CodeDeploy on Lambda	Canary (10% then the rest), Linear (10% every N minutes), AllAtOnce	"Shift traffic gradually, roll back on alarm"
ECS blue/green	CodeDeploy (canary, linear, all-at-once), or now native ECS blue/green with linear and canary	Rolling update is still the ECS default

Migration FOCUS

List	Members	Exam cue
7 Rs	Retire, Retain, Rehost, Relocate, Repurchase, Replatform, Refactor	Rehost: lift and shift (MGN). Relocate: VMware Cloud on AWS. Repurchase: move to SaaS. Replatform: e.g. to RDS
AWS Backup coverage	EC2, EBS, S3, RDS, Aurora, DynamoDB, EFS, FSx, Storage Gateway, DocumentDB, Neptune, Redshift, Timestream, EKS, SAP HANA on EC2, VMware VMs	"Central backup policy across accounts" means AWS Backup with Organizations

SOURCES

- SAP-C02 exam guide (domains and task statements)
- SAP-C02 In-Scope AWS Services
- AWS Snowball Edge availability change and device specifications
- AWS Snow device updates (Storage Blog)
- Transit Gateway quotas
- Lambda quotas
- SQS 1 MiB payload announcement
- API Gateway timeout beyond 29 s
- AWS Lifecycle Changes: Maintenance, Sunset, Full Shutdown
- Aurora Global Database: 10 secondary Regions
- Direct Connect features
- S3 maximum object size 50 TB
- RDS engines, Aurora DSQL, database decision guide, Timestream LiveAnalytics availability
- S3 storage classes, EBS volume types, FSx File Gateway
- Gateway endpoints, Route 53 routing policies, DX virtual interfaces
- EC2 instance type naming, burstable instances, ElastiCache engine comparison
- Placement groups, CodeDeploy deployment configurations, 7 Rs, AWS Backup feature availability
- AWS Organizations quotas and higher SCP quotas (May 2026)
- SQS message quotas, Kinesis quotas, EBS volume types, VPC quotas

AWS raises limits often (several changed in 2025 and 2026), so confirm any number before you rely on it in a real design.